

OM AVINASH PAILWAN

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Data Engineer with expertise in scalable ETL pipelines, Google Cloud Platform, and end-to-end data solutions. Proven track record of building data infrastructure while developing user-facing applications. Strong foundation in SQL, Python, GCP services, and Snowflake, with experience in frontend development for data visualization and monitoring dashboards. Proficient in using Cursor AI for accelerated development and have completed multiple AI-assisted projects

EDUCATION

Colorado State University | Colorado, USA

Master of Computer Information Systems
Graduate Teaching Assistant (ESS 555)

May 2025

GPA: 3.5/4.0

Pune University | Maharashtra, India

Bachelor of Computer Engineering Honors Data Science

June 2022

GPA: 3.5/4.0

SKILLS

Languages: SQL, Python, R, SAS, Java, Shell/UNIX, C#

Analytics: PySpark, Pandas, Jupyter, Power BI

Dataseystems: BigQuery, Teradata, MySQL, Snowflake, Oracle, Databricks, Spark, Kafka, Hadoop

Tools: Google Cloud Platform (CloudBuild, CloudRun, BigQuery, Kubernetes), AWS, Azure, DataStage, Airflow

Expertise: ETL/ELT, Data Modeling, Data Warehousing, Data Streaming, CI/CD for Data pipelines, Requirement gathering, Orchestration

WORK EXPERIENCE

Data Engineer | Oracle - India

Apr 2023-Jul 2024

- **Scalable ELT/ETL Development:** Designed and optimized enterprise-scale ELT/ETL processes with custom SQL/CCL transformations, processing daily data while reducing execution time by 20% and improving overall system efficiency by 55%
- **Enterprise Data Management:** Managed complex clinical data across multiple departments (**claims, revenue, nursing, radiology**), implementing robust data modeling and schema optimization that achieved 35% accuracy improvements and 100% system reliability
- **Cross-functional Collaboration:** Partnered with diverse stakeholders to deliver data warehouse solutions and reporting analytics, effectively translating complex technical concepts for both technical and business audiences while ensuring regulatory compliance
- **Performance Optimization:** Engineered custom data transformation processes improving overall efficiency by 55%, directly supporting business intelligence and reporting initiatives

Data Engineer | Datametica - India

Feb 2022-Mar 2023

- **Large-Scale Data Migration & Optimization:** Orchestrated strategic migration of 100+ **Teradata** data models to **BigQuery** cloud data warehouse, implementing efficient schemas and performance optimizations for enterprise-scale analytics workloads
- **Automated Pipeline & DevOps Development:** Built automated ELT/ETL **Airflow DAGs** using **Python**, while implementing CI/CD best practices that accelerated development **speed by 64%** through automated testing and deployment processes
- **Cloud Performance Optimization:** Optimized cloud data warehouse performance through strategic implementation of partitioned and clustered tables, achieving 60% query speed improvements for finance-critical analytical workloads
- **Architected a HIPAA-compliant healthcare data platform POC** leveraging Docker, Kubernetes, and GCP services (Cloud Run, Pub/Sub, VPC-SC, CMEK, audit logging) to securely process and analyze real-time audio data; implemented licensed California therapist verification for compliance and access control
- **Developed a real-time audio processing pipeline with AI integration** using Vertex AI sentiment analysis (5 themes, 3 sentiment levels) to extract clinical insights; designed microservices with CI/CD automation to minimize infrastructure costs and enable scalable deployment of AI services

PROJECT EXPERIENCE

- **Stock Market Analysis Using Kafka** ([GitHub - Stock-Market-Analysis-Using-kafka](#)) - Real-time streaming pipeline processing livestock market data with Kafka producers/consumers, Python analytics, and interactive dashboards for market trend monitoring and trading insights.
- **Streaming Data Pipeline for Real-Time Dashboard with Dataflow** ([GitHub - Streaming-Data-Pipeline-Real-Time-Dashboard-with-Dataflow](#)) - End-to-end streaming architecture using Google Cloud Dataflow for real-time data ingestion and transformation, powering live monitoring dashboards with automated alerting and performance metrics visualization.
- **Bank Data Analysis Using Databricks** ([GitHub - BankDataAnalysis-Using-Databricks](#)) - Comprehensive banking analytics platform leveraging Databricks for large-scale data processing, customer segmentation analysis, and financial risk assessment with automated ML pipelines and interactive visualizations.
- **Interactive Power BI Dashboard on Overstimulation Factors** ([GitHub - PowerBI-Overstimulation-Dashboard](#)) - Built a dynamic dashboard analyzing behavioral and lifestyle data using advanced DAX and visual interactivity to uncover insights into overstimulation contributors.

ACHIEVEMENTS/CERTIFICATIONS

- **Founded and led** a startup from concept to launch, ultimately sunsetting the venture after strategic evaluation of market fit and growth potential.
- **Certifications:** GCP [Associate Cloud Engineer Certification - Credly](#) and [Snowflake SnowPro Associate: Platform Certification](#)
- Awarded "[Rising Star Award](#)" for continued growth and success at Datametica,